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Cannabis and Mental Health
A Review

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IMPORTANCE Cannabis use is common among those with mental health conditions, and many people report using cannabis to manage mental health symptoms. It is important for clinicians to understand the lack of clear benefits of cannabis for mental health conditions and the potential for substantial adverse effects.

OBSERVATIONS Overall, the potential benefits of cannabis for mental health conditions remain poorly studied. There is low-certainty evidence that Δ-9-tetrahydrocannabinol (THC)–predominant cannabis may not improve symptoms of posttraumatic stress disorder, and there is largely insufficient evidence to characterize the effects of long-term THC-predominant cannabis use on anxiety, depression, and attention-deficit/hyperactivity disorder. There is emerging low-certainty evidence that the cannabis constituent cannabidiol alone may reduce anxiety in patients with anxiety disorders. THC-predominant cannabis use holds substantial risk for adverse mental health effects, and counseling patients about these risks is crucial to promote safety. These risks include worsening mania symptoms and function in those with bipolar disorder and an increase in psychotic symptoms in those with psychotic spectrum disorders. Among people with past-year cannabis use, about 3 in 10 have cannabis use disorder (CUD), and about one-half those with CUD have moderate or severe disease with negative social, employment, or other adverse outcomes. Regular use of high THC-content products by adolescents and young adults is associated with several concerning risks, including an increased risk of psychosis (estimates range from about 2-fold to 11-fold increased risk), a higher risk of CUD, and self-harm in those with mood disorders. Cannabis use should be avoided in individuals at elevated risk of harms, including adolescents and young adults, those with bipolar or psychotic disorders, pregnant individuals, and those at risk for substance use disorders.

CONCLUSIONS AND RELEVANCE The current evidence base is not sufficient to support the use of cannabis for the treatment of mental health conditions and demonstrates substantial risks of adverse effects. Clinicians should engage patients with mental health conditions in discussions about cannabis use because use is common, has an influence on mental health symptoms, and is likely an important modifiable risk factor for mental health conditions in some populations.

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Cannabis and products derived from cannabis have become more accessible, and their use—including daily or near-daily use—has grown dramatically over the last 2 decades.[1] Simultaneously, cannabis and cannabis derivatives have grown to be more potent.[2,3] These factors have synergized to establish cannabis use as a significant public health concern in the US.[4] Many people use cannabis for medical purposes: 10% to 30% of the general population and three-quarters of primary care patients who use cannabis report doing so to manage health-related symptoms.[5-9] As cannabis use becomes more prevalent, clinicians are frequently confronted with questions about cannabis and its implications for health during routine encounters.

The mental health effects of cannabis are complex, incompletely understood, and influenced by an array of substance-related and individual-level factors. Cannabis use, including heavy use, is common among those with preexisting mental health diagnoses and may represent an attempt to self-medicate conditions, such as posttraumatic stress disorder (PTSD), depression, and anxiety.[10] Further, cannabis can exacerbate or even cause significant psychiatric symptoms, as shown in a 2025 systematic review about the mental health effects of high-potency cannabis.[11] Even as cannabis potency and evidence about its potential adverse effects have steadily increased, the acceptance of and perception of risk associated with cannabis use has decreased.[12,13] Clinicians are uniquely positioned to counsel patients about the benefits and risks of cannabis; however, clinicians must gain knowledge and comfort with counseling patients about cannabis to fulfill this important role.

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Table 1. Glossary of Terms

Term	Definition
Cannabinoids	Chemical compounds that act on cannabinoid receptors; includes phytocannabinoids (plant-derived), endocannabinoids (body-made), and synthetic cannabinoids.
Cannabis	The plant (*Cannabis sativa*) that produces cannabinoids, including THC. Legally, plants with <0.3% THC can be called hemp.
CB1 receptor	A cannabinoid receptor highly expressed in the brain (cortex, hippocampus, basal ganglia); mediates psychoactive effects.
CB2 receptor	Found primarily in the immune system and peripheral tissues; involved in inflammation and immune modulation.
CBD	A nonpsychoactive compound with potential anxiolytic, antipsychotic, and anti-inflammatory effects.
CUD	A clinical diagnosis reflecting problematic cannabis use causing impairment or distress.
Endocannabinoids	Lipid-based neurotransmitters produced by the body that activate cannabinoid receptors (eg, anandamide, 2-acylglycerol); analogous to endorphins and opioid receptors.
Psychoactive	Producing changes in perception, mood, consciousness, or behavior (eg, THC is psychoactive, CBD is not).
Synthetic cannabinoids	Manufactured compounds that mimic THC (eg, JWH-018, Spice, and K2); often more potent and riskier.
THC	The main psychoactive compound in cannabis that produces intoxication.
Tolerance	Reduced response to a drug over time, requiring higher doses to achieve the same effect.
Withdrawal	Symptoms that occur when stopping or reducing cannabis after prolonged use (eg, irritability, insomnia).

Abbreviations: CB1, cannabinoid 1; CB2, cannabinoid 2; CBD, cannabidiol; CUD, cannabis use disorder; THC, Δ-9-tetrahydrocannabinol.

Patients report a preference for obtaining information about cannabis from health care professionals,[14] but most physicians report having limited knowledge and confidence in discussing cannabis with their patients.[15] While many reviews have examined the effects of cannabis on specific mental health conditions, few have synthesized evidence and clinical implications across conditions. The purpose of this article is to review the evidence on the effects of cannabis on commonly encountered mental health conditions, describe how components of cannabis differentially influence mental health, highlight groups at higher risk for mental health harms from cannabis use, and provide clinicians with basic information about cannabis pharmacology and biology. We hope to empower clinicians to hold evidence-informed discussions about cannabis and mental health.

Methods

Table 1 provides a glossary of terms used throughout this article, and **Table 2**[16-28] identifies the key evidentiary sources for each section of this review. The primary resource for this review is the Systematically Testing the Evidence on Marijuana project, which includes a variety of living systematic reviews (ie, updated annually in an ongoing basis) and other evidence resources examining the health effects of cannabis.[29] In particular, we used Systematically Testing the Evidence on Marijuana systematic reviews examining the effects of cannabis in patients with PTSD, anxiety, depression, and bipolar disorder as the basis for much of this review; because each of these reviews is updated regularly, we did not systematically search the literature beyond what was included in them.[16,17] These living systematic reviews adhere to contemporary methods, which have been previously posted, for synthesizing and characterizing the certainty of evidence across a body of literature and undergo annual literature surveillance.[30] We also include high-quality systematic reviews[31] examining the effects of cannabis on the risk and management of psychotic spectrum disorders[19-21] and summarize the current evidence for attention-deficit/hyperactivity disorder (ADHD) based on a recent scoping review.[22]

Review Results

Cannabis, Cannabinoids, and the Endocannabinoid System

Cannabis usually refers to the cannabis plant (*Cannabis sativa*) or plant-derived products, which encompass numerous formulations most often consumed through inhalation, orally, or topically (**Table 3**). Cannabis contains cannabinoids, terpenes, and flavonoids. The 2 best studied and most clinically relevant cannabinoids are Δ-9-tetrahydrocannabinol (THC) and cannabidiol (CBD). CBD, which has garnered substantial public interest, may have anxiolytic properties but is generally considered nonpsychoactive and remains poorly studied.[32] THC, the principal psychoactive constituent of cannabis, is a partial agonist at brain cannabinoid 1 receptors (CB1R)[33] and demonstrates a dose-response relationship between the amount of THC consumed and the type and intensity of psychoactive effect experienced (**Figure**).

Cannabis potency is determined by THC content and ranges considerably across cannabis formulations and derivatives.[3,34] Certain potent synthetic cannabinoids, such as illicitly manufactured Spice and K2, are high-affinity full agonists of CB1R and associated with a high risk of adverse mental health effects, such as psychosis.[35] It is important for clinicians to become familiar with cannabis formulations and potency because many of the mental health effects reviewed below may be closely related to THC content (Table 3).

The mental and physical health effects of cannabis are due to the interaction of exogenous cannabinoids (ie, from cannabis use) on the body's endocannabinoid system.[36,37] The endocannabinoid system is comprised of receptors, endocannabinoids, and their synthesizing/metabolizing enzymes. It is crucial for maintaining homeostasis across a broad variety of physiological processes in the brain and body.[36,37] The physiologic effects of endocannabinoids are typically local and rapidly inactivated.[37] In contrast, THC indiscriminately activates CB1Rs across the brain—including regions which govern mood, motivation, and memory—for long periods, giving rise to the intoxicating effects of cannabis.

Mental Health Effects of Cannabis and Cannabinoids

The following section reviews what is known about the potential therapeutic benefits and risks of cannabis use across a variety of common mental health conditions (Table 2).

PTSD

Cannabis is frequently used by people with PTSD to cope with symptoms such as hyperarousal and sleep problems.[38-40] The empirical literature examining the benefits and harms of cannabis for the treatment of PTSD symptoms remains underdeveloped. Overall, there is low-certainty evidence that cannabis does not reduce PTSD symptom severity or improve other mental health symptoms in patients with PTSD.[16]

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Table 2. Summary of Mental Health Effects of Cannabis and Cannabinoids

Condition	Cannabinoid(s) studied	Summary of evidence	Source
PTSD	THC, CBD, and nabilone	Cannabis did not reduce PTSD symptom severity or improve other mental health symptoms in patients with PTSD (low-certainty evidence)	Cannabis for the Management of Symptoms of PTSD[16]
Anxiety	• THC • CBD, 150-300 mg	• THC: Very uncertain • CBD: May reduce anxiety symptoms (low-certainty evidence)	Cannabis for Anxiety and Mood Disorders[17]
Depression	Unspecified	Very uncertain	Cannabis for Anxiety and Mood Disorders[17]
Bipolar disorder	• THC • CBD	• THC: Cannabis associated with worsened mania and decreased likelihood of recovery and remission (low-certainty evidence) • CBD: Very uncertain	Cannabis for Anxiety and Mood Disorders[17]
Psychosis	THC	• Heavy cannabis use (especially in adolescence) increased the risk of developing chronic psychotic disorders • Experiencing cannabis-induced psychosis also worsened this risk • Those with first-degree relatives with psychotic spectrum disorder were at higher risk • In those with established psychotic disorders, cannabis was associated with worse disease outcomes	• Petrilli et al,[18] 2022 • Marconi et al,[19] 2016 • Murrie et al,[20] 2020 • Schoeler et al,[21] 2016
ADHD	Balanced THC and CBD	1 Small trial showed no significant effects on ADHD symptoms	Francisco et al,[22] 2023
CUD	THC	• CUD prevalence of approximately 30% among people who use cannabis; higher risk for younger age, male sex, and those with SUD history • Cannabis withdrawal syndrome possible with cessation • Treatment options limited; some evidence for CBT, MET, and CM; no pharmacological options	• Prevalence and Risk of Developing Cannabis Use Disorder in Adults Who Use Cannabis[23] • VA/DoD Clinical Practice Guideline for the Management of Substance Use Disorders[24] • Pharmacotherapy for the Treatment of Cannabis Use Disorder[25]
Cognition	THC	• Acute intoxication is associated with mild to moderate deficits in verbal learning, working memory, executive functioning, and processing speed (depending on dose) • Adolescents with heavy or chronic use may experience slight declines in IQ; for adults, cognitive effects are likely to resolve with extended abstinence, although the effect of duration of use and abstinence on these improvements are not well understood	• Broyd et al,[26] 2016 • Dellazizzo et al,[27] 2022 • Power et al,[28] 2021

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; CBD, cannabidiol; CBT, cognitive behavioral therapy; CM, contingency management; CUD, cannabis use disorder; DoD, US Department of Defense; MET, motivational enhancement therapy; PTSD, posttraumatic stress disorder; SUD, substance use disorder; THC, Δ-9-tetrahydrocannabinol; VA, US Department of Veterans Affairs.

Table 3. Types of Cannabis Products and Modes of Use by Common Routes of Consumption

Characteristic	Route of consumption		
	Inhaled	Oral	Topical
Products	• Flower: dried cannabis buds for smoking/vaping (approximately 5%-35% THC) • Hash (hashish): pressed trichome resin; more potent than flower (approximately 30%-50% THC) • Concentrates: high-potency extracts (eg, shatter, wax, rosin, distillate; approximately 80%-90% THC)	• Hash • Concentrates	• Concentrates
Modes of use	• Joint: cannabis cigarette • Spliff: cannabis mixed with tobacco cigarette • Blunt: cannabis in a cigar/cigarillo wrap • Pipe (dry): cannabis in a pipe; not through water • Bong/water pipe: smoke pulled through water (cools/humidifies; does not remove most toxins) • Vape: heated flower or oil/concentrate to aerosolize cannabinoids without combustion • Dab/dabbing: high-cannabinoid concentrate applied to flame-heated glass or metal, then vaporized and inhaled; very high THC per hit	• Capsule: dronabinol, nabilone • Edibles: cannabis extract infusion, concentrate, or isolate incorporated into food or beverage products (baked goods, gummies, hard candy, butter) • Tincture/oil: typically, high-cannabinoid concentrate in alcohol, glycerin, or oil (eg, olive oil); can be administered sublingually	• Lotion, salve, balm, or patch: cannabis-infused carrier or concentrate preparation

Abbreviation: THC, Δ-9-tetrahydrocannabinol.

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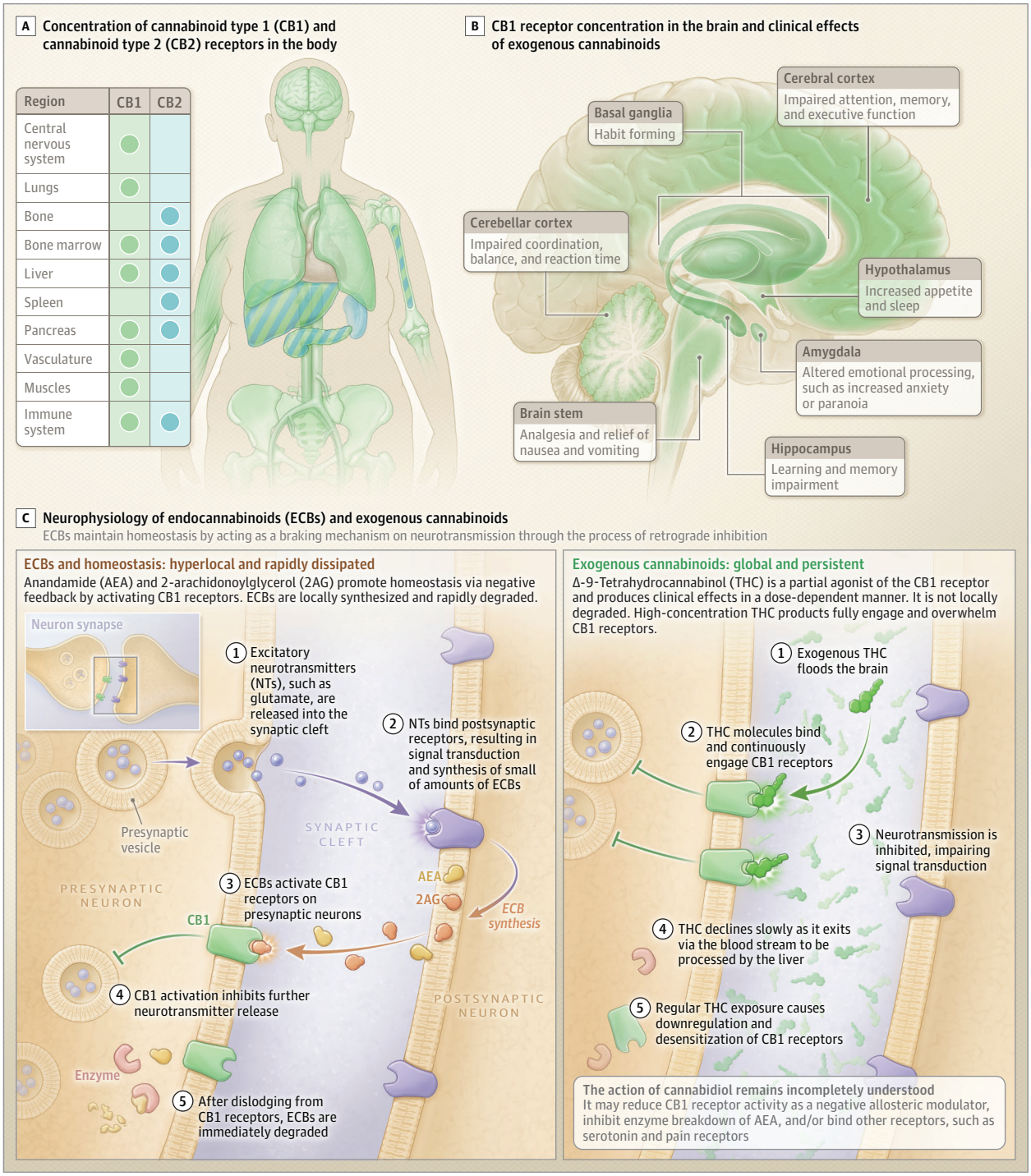
Figure. The Endocannabinoid System



To our knowledge, only 2 clinical trials have been published assessing cannabis for PTSD (although there are a number of ongoing trials, including several larger trials, due to be completed within the next 5 years).[16] The largest published trial enrolled 80 veterans with PTSD and used a crossover design to compare 3 different formulations of smoked cannabis and placebo; there were no differences in PTSD symptom severity between placebo and cannabis over 3 weeks of treatment.[41] There were also no differences in insomnia severity across groups in this trial.[41] Adverse effects, such as dry mouth, nausea, dizziness, and somnolence, were more common among participants taking THC-containing formulations than those taking CBD alone or placebo.[41] A small trial (N = 10)[42] found that nabilone (a synthetic form of THC) was associated with a reduction in nightmare severity but not with improvements in sleep quality; adverse effects were not reported. The small sample size and methodologic concerns temper confidence in this study's findings. Five

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observational studies have yielded mixed results, with some studies suggesting potential benefit, while others demonstrated potential harm.[16,43,44]

Anxiety

One-half of those who report the use of cannabis for symptom management do so to treat anxiety,[45] and chronic cannabis use has been associated with anxiety disorders,[46] but the experimental literature is limited. The therapeutic effects of cannabis on anxiety are also difficult to characterize concisely because effects may vary according to the specific cannabinoid and dose studied.

Acutely, THC has shown mixed effects on anxiety. A placebo-controlled double-blind study showed that a single dose of oral THC, 7.5 mg, reduced anxiety in some healthy volunteers, while a higher dose (THC, 12.5 mg) induced anxiety.[47] On the other hand, CBD likely has anxiolytic properties when used acutely; for example, several placebo-controlled trials have found 300 to 600 mg of CBD reduces acute anxiety from simulated public speaking in those with social phobia and in healthy volunteers.[48-50] Small studies in healthy volunteers suggest that CBD may also attenuate the anxiogenic effects of THC.[51,52] While the minor cannabinoid cannabigerol has garnered some interest for potential acute anxiolytic properties, future research is essential to verify a beneficial effect and to evaluate adverse effects.[53]

To date, treatment studies have provided insufficient evidence to draw conclusions about the long-term effects of THC-predominant cannabis use on anxiety. However, there is some early evidence suggesting that high-dose CBD may be beneficial for patients with anxiety disorders.[54] Low-certainty evidence from 3 trials and a quasi-experimental study have shown that daily CBD use (ranging from 150 to 300 mg) may reduce anxiety in patients with generalized anxiety disorder or social anxiety disorder over 4 to 15 weeks.[17] Importantly, there are several clinical trials currently underway assessing the efficacy of CBD for anxiety disorders that should help clarify its efficacy in the future.[17]

Depression

As with anxiety, cannabis is viewed by many in the general public as potentially beneficial for the relief of depression.[13] However, there is a dearth of clinical studies examining the efficacy of cannabis as a treatment for depression.[17] A meta-analysis of trials that examined depression symptoms as a secondary outcome in those with other primary diagnoses (usually multiple sclerosis) found cannabis was not associated with improved depression symptoms.[55] One prospective cohort study found that cannabis use was not associated with higher rates of remission among those with major depressive disorder.[56]

Observational studies suggest heavy cannabis use may be associated with self-harm. While it is unclear whether cannabis use is a form of self-medication and a marker for severe depression or is directly worsening depression symptoms, there are compelling reasons to raise concerns about chronic heavy use in those with poorly controlled depression, especially among younger populations. First, cannabis use has been associated with suicidality in young adults in a cross-sectional study,[57] and while the observational literature is limited by crude measures of exposure, heavier use does appear to be associated with a higher risk of suicidal behavior.[58] Second, in a retrospective cohort study among adolescents and young adults with mood disorders, cannabis use disorder (CUD) was associated with a higher risk of self-harm and death by unintentional opioid overdose.[59] Finally, a cross-sectional study found that veterans with CUD were at substantially higher risk of self-injurious behavior (both with and without suicidal intent).[60]

Bipolar Disorder

Consistent findings across 7 observational studies provide low-certainty evidence that cannabis use is associated with increased rates of mania and lower rates of recovery and remission among patients with bipolar disorder.[17] The EMBLEM study, which observed 3459 patients with bipolar disorder over 12 months, found that cannabis use was associated with higher rates of mania, higher rates of psychosis, lower rates of recovery, and more frequent adverse social and employment events.[61-64]

There is very little evidence examining the effects of CBD alone on outcomes in bipolar disorder.[17] One 2024 small, single-site trial[65] found CBD may not improve depression and anxiety outcomes in patients with bipolar disorder and acute depression, although there was a nonsignificant benefit observed with higher dose CBD (300 mg per day); a larger trial is currently being conducted.[66] Clinicians should counsel patients with bipolar disorder against the regular use of cannabis, especially high THC-content products, given the consistent evidence of increased rates of mania, poor outcomes, and the high morbidity and mortality associated with poorly controlled bipolar disorder.[67]

Psychosis

Experimental studies clearly show that THC can induce a transient psychotic state (typically lasting a few hours), while CBD does not.[68] However, in some instances, psychosis may last well beyond the period of acute intoxication; this cannabis-induced psychosis (CIP) often requires clinical intervention.[69] While an episode of CIP usually remits with abstinence from cannabis, a CIP episode may be a harbinger of later schizophrenia; up to 50% with CIP may go on to develop chronic psychotic disorders (eg, schizophrenia) with repeated use.[20,70,71] In individuals with established psychotic disorders, cannabis use has a negative effect on the course and expression of the illness, including more relapses, more positive symptoms, and more hospitalizations.[21,69]

The rates of psychosis associated with cannabis use are increasing[72]; these increases appear to be partly attributable to the availability of more potent cannabis products and are increasingly associated with CUD.[18,73-76] These trends are especially concerning for younger people. There is mounting epidemiological evidence that frequent exposure to cannabis (weekly or more frequent), particularly during adolescent periods of brain development, is associated with higher risk of developing schizophrenia or other psychosis-related symptoms.[19] For example, a 2024 longitudinal study[77] found an 11-fold increased risk of psychotic disorders associated with cannabis use during adolescence. Another prospective cohort study found that the risk of persistent psychotic symptoms in adulthood was greatest in those who started using cannabis during adolescence and then continued into adulthood.[78] A meta-analysis examining the relationship between cannabis use frequency and psychosis risk found a dose-response effect, with daily use conferring the greatest risk (daily use compared with no use: relative risk, 1.76; 95% CI, 1.47-2.12; weekly use

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compared with no use: relative risk, 1.35; 95% CI, 1.19-1.52).[79] Of note, the risk of experiencing psychotic symptoms in the context of cannabis use may be greater among those with a genetic predisposition to psychotic spectrum disorders.[80,81] Accordingly, caution against early-onset, heavy cannabis use—while important for all youth—may be particularly important for patients who have first-degree relatives with psychotic spectrum disorders.

CBD may have antipsychotic properties,[82,83] although current empiric evidence does not support its clinical use as an antipsychotic. People who use products high in CBD relative to THC content may have fewer psychotic experiences than people who use THC-predominant products.[84] However, in experimental studies, CBD has not consistently reduced the psychosis-inducing properties of THC in healthy volunteers or in those with psychotic spectrum disorders.[85,86]

ADHD

Cannabis use is common among those with ADHD,[87] and one-third to more than one-half of patients who use cannabis daily or seek treatment for CUD meet criteria for ADHD.[88,89] Cannabis is frequently used as a way to self-medicate ADHD symptoms.[87,89,90] However, acute cannabis intoxication may result in symptoms mimicking those of ADHD, including deficits in attention, working memory, and decision-making.[91] These deficits may persist beyond the acute intoxication phase in youth who use cannabis heavily.[27] To our knowledge, there has only been 1 small randomized clinical trial to date of an oromucosal spray containing similar amounts of THC and CBD in patients with ADHD, which found no statistically significant effects (positive or negative) on attention, memory, or cognition.[22,92]

CUD

CUD is characterized by impaired control of use, craving, social impairment, use in risky situations (eg, driving), tolerance, and withdrawal. The most recent estimates using the *Diagnostic and Statistical Manual of Mental Disorders* (Fifth Edition) definition for CUD find prevalence rates of about 30% among those who have used cannabis in the past year.[23] Younger individuals, male individuals, those with a history of substance use disorder, and those who use cannabis more frequently and at higher doses are at higher risk for CUD.[23] About one-half have moderate to severe disease, signifying that CUD contributes appreciably to negative social, employment, or other life impacts.[93] Evidence-based nonpharmacologic treatments include cognitive behavioral therapy, motivational enhancement therapy, and contingency management, although these are not readily accessible for many patients and may be of uncertain efficacy.[24] There are no US Food and Drug Administration–approved pharmacological treatments for CUD.[25]

Cannabis and Cognition

Acute intoxication with THC-predominant cannabis is associated in a dose-dependent manner with mild to moderate deficits in verbal learning, working memory, executive functioning, and processing speed.[26,91,94] Clinically, these types of deficits have been associated with driving impairment and motor vehicle accidents[95,96] and raise concern for safety-sensitive occupations.[97] While THC-predominant cannabis has acutely been shown to have dose-dependent deleterious cognitive effects, it is unclear whether cannabis doses used in therapeutic applications are associated with similar effects. For example, an oromucosal spray containing THC, 2.7 mg, and CBD, 2.5 mg, was not associated with impaired cognition in trials of patients with chronic pain.[98]

The chronic effects of cannabis use on cognition are complex and incompletely understood. Adolescents who use cannabis regularly may subsequently experience small declines in IQ, again underscoring concerns about regular use of THC-predominant cannabis in those with developing brains.[28] In adults, it is likely that cognitive effects improve with abstinence, although it is unclear how duration of use and duration of abstinence modifies these improvements.[27,91,94]

Table 4. Cannabinoid Interactions With Psychoactive Medications

Medication	Cannabis/THC/CBD effect on the drug
Antidepressants	
Amitriptyline, desipramine, and imipramine	Inhibits cytochrome P450 2C19 and cytochrome P450 2D6, increases drug level; drug and THC both increase anticholinergic effects (eg, tachycardia)
Citalopram and escitalopram	Increase adverse effects, presumably via cytochrome P450 2C19 inhibition
Antiepileptics	
Carbamazepine	Inhibits metabolism and increases drug level
Clobazam	CBD inhibits cytochrome P450 2C19, increases drug level
Valproic acid	Elevated liver enzymes, variable change in drug level
Benzodiazepines	
Clonazepam, diazepam, and lorazepam	Both THC and CBD inhibit cytochrome P450 3A4, while CBD inhibits UDP-glucuronosyltransferase 2B7, increases drug level and risk of sedation
Opioids	
Buprenorphine	Inhibits cytochrome P450 3A4 and increases drug levels

Abbreviations. CBD, cannabidiol; THC, Δ-9-tetrahydrocannabinol.

Cannabinoid Interactions With Mental Health Medications

THC and CBD are substrates, inhibitors, and inducers of multiple metabolic enzymes, which may lead to consequential drug-drug interactions with several commonly prescribed medications.[99] Drug interaction checkers specific to cannabis are publicly available (eg, Penn State CANN-DIR Project[100]), and clinicians should check for interactions with a patient's medications as a part of counseling around cannabis use. A summary of interactions with select psychiatric medications is available in **Table 4**. For example, CBD can increase concentrations of the antiepileptic clobazam which could increase somnolence and ataxia.[101] CBD may also have led to increased cough, diarrhea, dizziness, and fatigue experienced by adolescents taking certain selective serotonin reuptake inhibitors.[102] Both THC and CBD can interfere with the metabolism of tricyclic antidepressants, and anticholinergic effects from both THC and tricyclic antidepressants may increase the experience of tachycardia and delirium.[95,103] The inhibition of cytochrome P450 3A4 may be responsible for higher buprenorphine and norbuprenorphine serum concentrations in people who use cannabis compared with those who do not,[104] suggesting the need for monitoring of opioid withdrawal if patients taking buprenorphine reduce cannabis use. Levels of the antiepileptic and mood stabilizer carbamazepine have been reported to fluctuate with cannabis use,[105] and elevated liver enzymes (which can also result from CBD alone)[106] with coadministered valproate have been reported in multiple studies.[107]

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Cannabis has the potential to amplify the sedating effects of benzodiazepines, both because of decreased metabolism and because both drugs can cause sedation through different mechanisms. Both THC and CBD can increase concentrations of clonazepam and diazepam, while CBD can increase the concentration lorazepam.[100]

Talking With Patients About Cannabis Use and Mental Health

Clinicians are encouraged to become comfortable discussing the benefits and risks of cannabis, as they will care for patients who use—or are interested in using—cannabis to manage mental health symptoms or those with a mental health diagnosis who are using cannabis recreationally. Most of the US population now lives in states where cannabis can be purchased for recreational use or for medical use with a qualifying condition.[108] In practice, patients are getting advice about cannabis from dispensary workers, most of whom lack medical training and typically do not engage in the kind of comprehensive discussion of risks and benefits that a clinician would.[109]

One of the challenges in translating evidence about cannabis into clinical practice is the frequent misalignment between the published evidence, current medical cannabis policies, and patient experience. The current evidence base largely does not support the use of cannabis to treat mental health conditions, but clinicians are likely to encounter patients whose anecdotal experience is positive. When faced with therapeutic uncertainty, there is still some advice clinicians can offer to minimize harms associated with cannabis use without dismissing patient experience. Many of these suggestions draw on clinical guidance statements about the use cannabis for chronic pain[110-112] and the Lower Risk Cannabis Use Guidelines, which provides important harm reduction advice.[113]

Ask About Cannabis Use

Screening for substance use, including cannabis use, is recommended by the US Preventive Services Task Force, but cannabis use screening in primary care has not yet been widely adopted.[114] Several screening tools to detect problematic use have been evaluated for use in primary care settings, including the Single Item Screen-Cannabis,[115] which has a high negative predictive value and is useful for ruling out CUD. However, given its low positive predictive value, positive screens should be followed by a more detailed assessment, such as the 8-item Cannabis Use Disorder Identification Test–Revised,[116] which performs reasonably well in primary care settings. Regardless of the tool used, clinicians should invite discussion about cannabis health effects by building a habit of asking about use in a routine and nonjudgmental manner.

Discuss Mental Health Harms Regarding THC Content and Frequency of Cannabis Use

Both acute and longer-term anxiogenic and psychosis-inducing properties of cannabis are likely related to THC content. Patients should generally be encouraged to choose lower-potency (ie, lower THC content) products (Figure). Some high-potency formulations, such as concentrates and dabs, should be avoided altogether, as there is no known therapeutic benefit to justify their inherent risk. Similarly, unregulated synthetic cannabinoids can be highly potent, contain toxic adulterants, and should be avoided altogether.

Clinicians can help patients minimize the risk for adverse events by teaching them to quantify and track THC dose. Methods for doing so are not yet well validated but can be useful to approximate individual use. For smoked cannabis products, multiply the weight of the product used in milligrams over a given time interval by the THC concentration to calculate the approximate amount of THC consumed (researchers have proposed correction factors to account for variable inhalation of smoked products, but these have not been validated for clinical use).[117] Edible and drinkable products purchased in dispensaries are typically labeled with THC content per serving. The median achieved THC dose (after titration) in trials of cannabis and cannabinoids for the management of chronic pain is 20 to 25 mg, although patients naive to cannabis should start at much lower doses of 2 to 5 mg per day.[118] An expert-based consensus cannabis dosing protocol for patients with chronic pain proposed a ceiling dose of 40 mg per day.[110] By comparison, in a 2025 study of people using cannabis daily, the median daily dose of THC was 130 mg.[34]

Discuss Mental Health Harms in Subgroups of Patients Who Should Be Discouraged From Using Cannabis

Adolescents and young adults are at greater risk for negative outcomes from frequent cannabis use. Regular (ie, more than weekly) THC use by those with developing brains can increase the risk of developing psychotic spectrum disorders; this risk may be even greater in those who have first-degree relatives with psychotic spectrum disorders. For these reasons, cannabis use should be discouraged in adolescents and young adults. Cannabis use during pregnancy should also be discouraged because there is moderate-certainty evidence that it increases the risk of adverse neonatal outcomes, such as preterm birth and low birth weight.[119,120]

Additionally, cannabis should be avoided in patients with certain underlying mental health conditions, including bipolar or psychotic spectrum disorders. While not well studied, patients with poorly controlled depression and anxiety should probably avoid cannabis use because it may increase symptom burden and because heavier use has been associated with self-harm in younger populations with mood disorders.[57,59] Patients with a history of substance use disorders should be counseled that they have higher risk of developing CUD.[23]

Be Aware of Cannabis Withdrawal and the Cycle of Use to Prevent Withdrawal Symptoms

People who use cannabis regularly for an extended time then stop use can develop a withdrawal syndrome (usually starting within 1 to 2 days, peaking after 2 to 6 days).[121] Patients should be made aware of cannabis withdrawal symptoms—insomnia, anxiety, restlessness, and agitation—so they are not perpetuating problematic use by mistakenly treating withdrawal symptoms as primary mental health symptoms.[122,123] There are tools such as the Marijuana Withdrawal Checklist and the Cannabis Withdrawal Scale to assess cannabis withdrawal, but none have been well validated.[121-123] Slow tapering could minimize withdrawal symptoms. There is low- and very low-certainty evidence that CBD and gabapentin, respectively, may improve withdrawal symptoms.[25]

Avoid Smoking as a Long-Term Route of Administration

While the long-term effects of cannabis smoking are not well studied, smoking is probably best avoided as a long-term route of administration for cannabis because of its propensity to increase rates

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of bronchitis, the potential to increase the risk of chronic lung disease, and the carcinogenic risk of cannabis smoke with potential for increased risk of lung and other cancers.[124-126]

Conclusions

Cannabis and its derivatives are commonly used by those with mental health symptoms and conditions. There are no therapeutic uses of cannabis for mental health conditions that are currently supported by the empiric literature. However, there are striking evidence gaps, and future research may help clarify the role of cannabis in managing mental health conditions. Moreover, clinicians must be aware that cannabis use—particularly regular use of higher THC doses—can pose substantial risks to mental health, especially in vulnerable groups, such as adolescents and young adults with developing brains; those with poorly controlled mental health symptoms, bipolar disorder, or psychotic disorders; those at high risk for psychotic disorders (with prior psychosis events or family history); pregnant individuals; and those at risk for substance use disorders. Clinicians have a vital role to play in educating and counseling patients about the use of cannabis to address or treat mental health symptoms and conditions and about the potential for serious adverse effects.

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REFERENCES

1. Caulkins JP. Changes in self-reported cannabis use in the United States from 1979 to 2022. *Addiction*. 2024;119(9):1648-1652. doi:10.1111/add.16519

2. ElSohly MA, Chandra S, Radwan M, Majumdar CG, Church JC. A comprehensive review of cannabis potency in the United States in the last decade. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2021;6(6):603-606. doi:10.1016/j.bpsc.2020.12.016

3. Freeman TP, Craft S, Wilson J, et al. Changes in delta-9-tetrahydrocannabinol (THC) and cannabidiol (CBD) concentrations in cannabis over time: systematic review and meta-analysis. *Addiction*. 2021;116(5):1000-1010. doi:10.1111/add.15253

4. National Academies of Sciences, Engineering, and Medicine; Teutsch SM, Hurd YL, Boyle EB, eds. *Cannabis Policy Impacts Public Health and Health Equity*. The National Academies Press; 2024.

5. Bonn-Miller MO, Boden MT, Bucossi MM, Babson KA. Self-reported cannabis use characteristics, patterns and helpfulness among medical cannabis users. *Am J Drug Alcohol Abuse*. 2014;40(1):23-30. doi:10.3109/00952990.2013.821477

6. Shi Y. Heterogeneities in administration methods among cannabis users by use purpose and state legalization status: findings from a nationally representative survey in the United States, 2020. *Addiction*. 2021;116(7):1782-1793. doi:10.1111/add.15342

7. Schauer GL, Roehler DR, Hoots BE. Self-reported medical and nonmedical cannabis use: results from the 2018 Behavioral Risk Factor Surveillance System in 14 states. *J Community Health*. 2022;47(4):641-649. doi:10.1007/s10900-021-01049-1

8. Gelberg L, Beck D, Koerber J, et al. Cannabis use reported by patients receiving primary care in a large health system. *JAMA Netw Open*. 2024;7(6):e2414809. doi:10.1001/jamanetworkopen.2024.14809

9. Terry-McElrath YM, Patrick ME. Medical cannabis use across ages 19-65 years: U.S. young and middle adults, 2018-2023. *Am J Prev Med*. 2025;70(3):107994. doi:10.1016/j.amepre.2025.107994

10. Rup J, Freeman TP, Perlman C, Hammond D. Cannabis and mental health: prevalence of use and modes of cannabis administration by mental health status. *Addict Behav*. 2021;121:106991. doi:10.1016/j.addbeh.2021.106991

11. Rittiphairoj T, Leslie L, Oberste JP, et al. High-concentration delta-9-tetrahydrocannabinol cannabis products and mental health outcomes: a systematic review. *Ann Intern Med*. 2025;178(10):1429-1440. doi:10.7326/ANNALS-24-03819

12. Compton WM, Han B, Jones CM, Blanco C, Hughes A. Marijuana use and use disorders in adults in the USA, 2002-14: analysis of annual cross-sectional surveys. *Lancet Psychiatry*. 2016;3(10):954-964. doi:10.1016/S2215-0366(16)30208-5

13. Keyhani S, Steigerwald S, Ishida J, et al. Risks and benefits of marijuana use: a national survey of U.S. adults. *Ann Intern Med*. 2018;169(5):282-290. doi:10.7326/M18-0810

14. Pergam SA, Woodfield MC, Lee CM, et al. Cannabis use among patients at a comprehensive cancer center in a state with legalized medicinal and recreational use. *Cancer*. 2017;123(22):4488-4497. doi:10.1002/cncr.30879

15. Rønne ST, Rosenbæk F, Pedersen LB, et al. Physicians' experiences, attitudes, and beliefs towards medical cannabis: a systematic literature review. *BMC Fam Pract*. 2021;22(1):212. doi:10.1186/s12875-021-01559-w

16. Ayers C, Harrod C, Durbin S, et al. Cannabis for the management of symptoms of PTSD: a living systematic review. Accessed August 11, 2025. https://www.cannabisevidence.org/evidence-syntheses/ptsd/

17. Ayers C, Shaw B, Robalino R, et al. Cannabis for anxiety and mood disorders. Accessed August 11, 2025. https://www.cannabisevidence.org/evidence-syntheses/anxiety_and_mood_disorders/

18. Petrilli K, Ofori S, Hines L, Taylor G, Adams S, Freeman TP. Association of cannabis potency with mental ill health and addiction: a systematic review. *Lancet Psychiatry*. 2022;9(9):736-750. doi:10.1016/S2215-0366(22)00161-4

19. Marconi A, Di Forti M, Lewis CM, Murray RM, Vassos E. Meta-analysis of the association between the level of cannabis use and risk of psychosis. *Schizophr Bull*. 2016;42(5):1262-1269. doi:10.1093/schbul/sbw003

20. Murrie B, Lappin J, Large M, Sara G. Transition of substance-induced, brief, and atypical psychoses to schizophrenia: a systematic review and meta-analysis. *Schizophr Bull*. 2020;46(3):505-516. doi:10.1093/schbul/sbz102

21. Schoeler T, Monk A, Sami MB, et al. Continued versus discontinued cannabis use in patients with psychosis: a systematic review and meta-analysis. *Lancet Psychiatry*. 2016;3(3):215-225. doi:10.1016/S2215-0366(15)00363-6

22. Francisco AP, Lethbridge G, Patterson B, Goldman Bergmann C, Van Ameringen M. Cannabis use in attention - deficit/hyperactivity disorder (ADHD): a scoping review. *J Psychiatr Res*. 2023;157:239-256. doi:10.1016/j.jpsychires.2022.11.029

23. Ayers C, Robalino S, Shaw B, Ward R, Kansagara D. Prevalence and risk of developing cannabis use disorder in adults who use cannabis. Accessed May 14, 2025. https://www.cannabisevidence.org/evidence-syntheses/adult-cannabis-use-disorder/

24. Department of Veterans Affairs, Department of Defense. VA/DoD clinical practice guideline for the management of substance use disorders. Accessed July 10, 2025. https://www.healthquality.va.gov/guidelines/MH/sud/VA-DoD-SUD-CPG_Final_for-508_v3.pdf

25. Durbin S, Ayers C, Yeddala S, Shaw B, Robalino R, Kansagara D. Pharmacotherapy for the treatment of cannabis use disorder. Accessed July 10, 2025. https://

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www.cannabisevidence.org/evidence-syntheses/pharmaceuticals-for-cannabis-use-disorder/

26. Broyd SJ, van Hell HH, Beale C, Yücel M, Solowij N. Acute and chronic effects of cannabinoids on human cognition-a systematic review. *Biol Psychiatry*. 2016;79(7):557-567. doi:10.1016/j.biopsych.2015.12.002

27. Dellazizzo L, Potvin S, Giguère S, Dumais A. Evidence on the acute and residual neurocognitive effects of cannabis use in adolescents and adults: a systematic meta-review of meta-analyses. *Addiction*. 2022;117(7):1857-1870. doi:10.1111/add.15764

28. Power E, Sabherwal S, Healy C, O' Neill A, Cotter D, Cannon M. Intelligence quotient decline following frequent or dependent cannabis use in youth: a systematic review and meta-analysis of longitudinal studies. *Psychol Med*. 2021;51(2):194-200. doi:10.1017/S0033291720005036

29. Systematically Testing the Evidence on Marijuana. About. Accessed August 8, 2025. https://www.cannabisevidence.org/about/

30. Systematically Testing the Evidence on Marijuana. Living systematic review methods. Accessed October 9, 2025. https://www.cannabisevidence.org/about/living-systematic-review-methods/

31. Shea BJ, Reeves BC, Wells G, et al. AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. *BMJ*. 2017;358:j4008. doi:10.1136/bmj.j4008

32. Freeman AM, Petrilli K, Lees R, et al. How does cannabidiol (CBD) influence the acute effects of delta-9-tetrahydrocannabinol (THC) in humans? a systematic review. *Neurosci Biobehav Rev*. 2019;107:696-712. doi:10.1016/j.neubiorev.2019.09.036

33. Stella N. THC and CBD: similarities and differences between siblings. *Neuron*. 2023;111(3):302-327. doi:10.1016/j.neuron.2022.12.022

34. Borodovsky JT, Hasin DS, Wall M, et al. Quantity of delta-9-tetrahydrocannabinol consumption and cannabis use disorder among daily cannabis consumers. *Addiction*. 2025;120(4):676-685. doi:10.1111/add.16700

35. Castaneto MS, Gorelick DA, Desrosiers NA, Hartman RL, Pirard S, Huestis MA. Synthetic cannabinoids: epidemiology, pharmacodynamics, and clinical implications. *Drug Alcohol Depend*. 2014;144:12-41. doi:10.1016/j.drugalcdep.2014.08.005

36. Mechoulam R, Parker LA. The endocannabinoid system and the brain. *Annu Rev Psychol*. 2013;64:21-47. doi:10.1146/annurev-psych-113011-143739

37. Lu HC, Mackie K. Review of the endocannabinoid system. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2021;6(6):607-615. doi:10.1016/j.bpsc.2020.07.016

38. Bonn-Miller MO, Babson KA, Vandrey R. Using cannabis to help you sleep: heightened frequency of medical cannabis use among those with PTSD. *Drug Alcohol Depend*. 2014;136:162-165. doi:10.1016/j.drugalcdep.2013.12.008

39. Boehnke KF, Sinclair R, Gordon F, et al. Trends in U.S. medical cannabis registrations, authorizing clinicians, and reasons for use from 2020 to 2022. *Ann Intern Med*. 2024;177(4):458-466. doi:10.7326/M23-2811

40. Loflin M, Earleywine M, Bonn-Miller M. Medicinal versus recreational cannabis use: patterns of cannabis use, alcohol use, and cued-arousal among veterans who screen positive for PTSD. *Addict Behav*. 2017;68:18-23. doi:10.1016/j.addbeh.2017.01.008

41. Bonn-Miller MO, Sisley S, Riggs P, et al. The short-term impact of 3 smoked cannabis preparations versus placebo on PTSD symptoms: a randomized cross-over clinical trial. *PLoS One*. 2021;16(3):e0246990. doi:10.1371/journal.pone.0246990

42. Jetly R, Heber A, Fraser G, Boisvert D. The efficacy of nabilone, a synthetic cannabinoid, in the treatment of PTSD-associated nightmares: a preliminary randomized, double-blind, placebo-controlled cross-over design study. *Psychoneuroendocrinology*. 2015;51:585-588. doi:10.1016/j.psyneuen.2014.11.002

43. Bonn-Miller MO, Brunstetter M, Simonian A, et al. The long-term, prospective, therapeutic impact of cannabis on post-traumatic stress disorder. *Cannabis Cannabinoid Res*. 2022;7(2):214-223. doi:10.1089/can.2020.0056

44. Wilkinson ST, Stefanovics E, Rosenheck RA. Marijuana use is associated with worse outcomes in symptom severity and violent behavior in patients with posttraumatic stress disorder. *J Clin Psychiatry*. 2015;76(9):1174-1180. doi:10.4088/JCP.14m09475

45. Kosiba JD, Maisto SA, Ditre JW. Patient-reported use of medical cannabis for pain, anxiety, and depression symptoms: systematic review and meta-analysis. *Soc Sci Med*. 2019;233:181-192. doi:10.1016/j.socscimed.2019.06.005

46. Beletsky A, Liu C, Lochte B, Samuel N, Grant I. Cannabis and anxiety: a critical review. *Med Cannabis Cannabinoids*. 2024;7(1):19-30. doi:10.1159/000534855

47. Childs E, Lutz JA, de Wit H. Dose-related effects of delta-9-THC on emotional responses to acute psychosocial stress. *Drug Alcohol Depend*. 2017;177:136-144. doi:10.1016/j.drugalcdep.2017.03.030

48. Bergamaschi MM, Queiroz RH, Chagas MH, et al. Cannabidiol reduces the anxiety induced by simulated public speaking in treatment-naïve social phobia patients. *Neuropsychopharmacology*. 2011;36(6):1219-1226. doi:10.1038/npp.2011.6

49. Linares IM, Zuardi AW, Pereira LC, et al. Cannabidiol presents an inverted U-shaped dose-response curve in a simulated public speaking test. *Braz J Psychiatry*. 2019;41(1):9-14. doi:10.1590/1516-4446-2017-0015

50. Zuardi AW, Cosme RA, Graeff FG, Guimarães FS. Effects of ipsapirone and cannabidiol on human experimental anxiety. *J Psychopharmacol*. 1993;7(1)(suppl):82-88. doi:10.1177/026988119300700112

51. Karniol IG, Shirakawa I, Kasinski N, Pfeferman A, Carlini EA. Cannabidiol interferes with the effects of delta 9 - tetrahydrocannabinol in man. *Eur J Pharmacol*. 1974;28(1):172-177. doi:10.1016/0014-2999(74)90129-0

52. Zuardi AW, Shirakawa I, Finkelfarb E, Karniol IG. Action of cannabidiol on the anxiety and other effects produced by delta 9-THC in normal subjects. *Psychopharmacology (Berl)*. 1982;76(3):245-250. doi:10.1007/BF00432554

53. Cuttler C, Stueber A, Cooper ZD, Russo E. Acute effects of cannabigerol on anxiety, stress, and mood: a double-blind, placebo-controlled, crossover, field trial. *Sci Rep*. 2024;14(1):16163. doi:10.1038/s41598-024-66879-0

54. Han K, Wang JY, Wang PY, Peng YC. Therapeutic potential of cannabidiol (CBD) in anxiety disorders: a systematic review and meta-analysis. *Psychiatry Res*. 2024;339:116049. doi:10.1016/j.psychres.2024.116049

55. Black N, Stockings E, Campbell G, et al. Cannabinoids for the treatment of mental disorders and symptoms of mental disorders: a systematic review and meta-analysis. *Lancet Psychiatry*. 2019;6(12):995-1010. doi:10.1016/S2215-0366(19)30401-8

56. Feingold D, Rehm J, Lev-Ran S. Cannabis use and the course and outcome of major depressive disorder: a population based longitudinal study. *Psychiatry Res*. 2017;251:225-234. doi:10.1016/j.psychres.2017.02.027

57. Han B, Compton WM, Einstein EB, Volkow ND. Associations of suicidality trends with cannabis use as a function of sex and depression status. *JAMA Netw Open*. 2021;4(6):e2113025. doi:10.1001/jamanetworkopen.2021.13025

58. Borges G, Bagge CL, Orozco R. A literature review and meta-analyses of cannabis use and suicidality. *J Affect Disord*. 2016;195:63-74. doi:10.1016/j.jad.2016.02.007

59. Fontanella CA, Steelesmith DL, Brock G, Bridge JA, Campo JV, Fristad MA. Association of cannabis use with self-harm and mortality risk among youths with mood disorders. *JAMA Pediatr*. 2021;175(4):377-384. doi:10.1001/jamapediatrics.2020.5494

60. Kimbrel NA, Meyer EC, DeBeer BB, Gulliver SB, Morissette SB. The impact of cannabis use disorder on suicidal and nonsuicidal self-injury in Iraq/Afghanistan-era veterans with and without mental health disorders. *Suicide Life Threat Behav*. 2018;48(2):140-148. doi:10.1111/sltb.12345

61. Zorrilla I, Aguado J, Haro JM, et al. Cannabis and bipolar disorder: does quitting cannabis use during manic/mixed episode improve clinical/functional outcomes? *Acta Psychiatr Scand*. 2015;131(2):100-110. doi:10.1111/acps.12366

62. Vieta E, Angst J, Reed C, Bertsch J, Haro JM; EMBLEM Advisory Board. Predictors of switching from mania to depression in a large observational study across Europe (EMBLEM). *J Affect Disord*. 2009;118(1-3):118-123. doi:10.1016/j.jad.2009.02.007

63. van Rossum I, Boomsma M, Tenback D, Reed C, van Os J; EMBLEM Advisory Board. Does cannabis use affect treatment outcome in bipolar disorder? a longitudinal analysis. *J Nerv Ment Dis*. 2009;197(1):35-40. doi:10.1097/NMD.0b013e31819292a6

64. Haro JM, van Os J, Vieta E, Reed C, Lorenzo M, Goetz I; EMBLEM Advisory Board. Evidence for three distinct classes of 'typical', 'psychotic' and 'dual' mania: results from the EMBLEM study. *Acta Psychiatr Scand*. 2006;113(2):112-120. doi:10.1111/j.1600-0447.2005.00692.x

65. Pinto JV, Crippa JAS, Ceresér KM, et al. Cannabidiol as an adjunctive treatment for acute bipolar depression: a pilot study: le cannabidiol comme traitement d'appoint de la dépression bipolaire aiguë: une étude pilote. *Can J Psychiatry*. 2024;69(4):242-251. doi:10.1177/07067437231209650

66. Cannabidiol for Bipolar Depression (CBD-BD) (CBD-BD). ClinicalTrials.gov identifier: NCT05867849. Updated April 23, 2025.

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Accessed June 6, 2023. https://clinicaltrials.gov/ct2/show/NCT05867849

67. Paljärvi T, Herttua K, Taipale H, et al. Cause-specific excess mortality after first diagnosis of bipolar disorder: population-based cohort study. *BMJ Ment Health*. 2023;26(1):e300700. doi:10.1136/bmjment-2023-300700

68. Hindley G, Beck K, Borgan F, et al. Psychiatric symptoms caused by cannabis constituents: a systematic review and meta-analysis. *Lancet Psychiatry*. 2020;7(4):344-353. doi:10.1016/S2215-0366(20)30074-2

69. D'Souza DC, DiForti M, Ganesh S, et al. Consensus paper of the WFSBP task force on cannabis, cannabinoids and psychosis. *World J Biol Psychiatry*. 2022;23(10):719-742. doi:10.1080/15622975.2022.2038797

70. Starzer MSK, Nordentoft M, Hjorthøj C. Rates and predictors of conversion to schizophrenia or bipolar disorder following substance-induced psychosis. *Am J Psychiatry*. 2018;175(4):343-350. doi:10.1176/appi.ajp.2017.17020223

71. Arendt M, Rosenberg R, Foldager L, Perto G, Munk-Jørgensen P. Cannabis-induced psychosis and subsequent schizophrenia-spectrum disorders: follow-up study of 535 incident cases. *Br J Psychiatry*. 2005;187:510-515. doi:10.1192/bjp.187.6.510

72. Hjorthøj C, Larsen MO, Starzer MSK, Nordentoft M. Annual incidence of cannabis-induced psychosis, other substance-induced psychoses and dually diagnosed schizophrenia and cannabis use disorder in Denmark from 1994 to 2016. *Psychol Med*. 2021;51(4):617-622. doi:10.1017/S0033291719003532

73. van der Steur SJ, Batalla A, Bossong MG. Factors moderating the association between cannabis use and psychosis risk: a systematic review. *Brain Sci*. 2020;10(2):97. doi:10.3390/brainsci10020097

74. Myran DT, Pugliese M, Harrison LD, et al. Changes in incident schizophrenia diagnoses associated with cannabis use disorder after cannabis legalization. *JAMA Netw Open*. 2025;8(2):e2457868. doi:10.1001/jamanetworkopen.2024.57868

75. Di Forti M, Marconi A, Carra E, et al. Proportion of patients in south London with first-episode psychosis attributable to use of high potency cannabis: a case-control study. *Lancet Psychiatry*. 2015;2(3):233-238. doi:10.1016/S2215-0366(14)00117-5

76. Di Forti M, Quattrone D, Freeman TP, et al; EU-GEI WP2 Group. The contribution of cannabis use to variation in the incidence of psychotic disorder across Europe (EU-GEI): a multicentre case-control study. *Lancet Psychiatry*. 2019;6(5):427-436. doi:10.1016/S2215-0366(19)30048-3

77. McDonald AJ, Kurdyak P, Rehm J, Roerecke M, Bondy SJ. Age-dependent association of cannabis use with risk of psychotic disorder. *Psychol Med*. 2024;54(11):2926-2936. doi:10.1017/S0033291724000990

78. Kuepper R, van Os J, Lieb R, Wittchen HU, Höfler M, Henquet C. Continued cannabis use and risk of incidence and persistence of psychotic symptoms: 10 year follow-up cohort study. *BMJ*. 2011;342:d738. doi:10.1136/bmj.d738

79. Robinson T, Ali MU, Easterbrook B, Hall W, Jutras-Aswad D, Fischer B. Risk-thresholds for the association between frequency of cannabis use and the development of psychosis: a systematic review and meta-analysis. *Psychol Med*. 2023;53(9):3858-3868. doi:10.1017/S0033291722000502

80. Wainberg M, Jacobs GR, di Forti M, Tripathy SJ. Cannabis, schizophrenia genetic risk, and psychotic experiences: a cross-sectional study of 109,308 participants from the UK Biobank. *Transl Psychiatry*. 2021;11(1):211. doi:10.1038/s41398-021-01330-w

81. Johnson EC, Colbert SMC, Jeffries PW, et al. Associations between cannabis use, polygenic liability for schizophrenia, and cannabis-related experiences in a sample of cannabis users. *Schizophr Bull*. 2023;49(3):778-787. doi:10.1093/schbul/sbac196

82. Bhattacharyya S, Wilson R, Appiah-Kusi E, et al. Effect of cannabidiol on medial temporal, midbrain, and striatal dysfunction in people at clinical high risk of psychosis: a randomized clinical trial. *JAMA Psychiatry*. 2018;75(11):1107-1117. doi:10.1001/jamapsychiatry.2018.2309

83. Bhattacharyya S, Morrison PD, Fusar-Poli P, et al. Opposite effects of delta-9-tetrahydrocannabinol and cannabidiol on human brain function and psychopathology. *Neuropsychopharmacology*. 2010;35(3):764-774. doi:10.1038/npp.2009.184

84. Schubart CD, Sommer IE, van Gastel WA, Goetgebuer RL, Kahn RS, Boks MP. Cannabis with high cannabidiol content is associated with fewer psychotic experiences. *Schizophr Res*. 2011;130(1-3):216-221. doi:10.1016/j.schres.2011.04.017

85. Englund A, Oliver D, Chesney E, et al. Does cannabidiol make cannabis safer? a randomised, double-blind, cross-over trial of cannabis with four different CBD:THC ratios. *Neuropsychopharmacology*. 2023;48(6):869-876. doi:10.1038/s41386-022-01478-z

86. Chesney E, Oliver D, Sarma A, et al. Does cannabidiol reduce the adverse effects of cannabis in schizophrenia? a randomised, double-blind, cross-over trial. *Neuropsychopharmacology*. 2025;50(12):1759-1767. doi:10.1038/s41386-025-02175-3

87. Ryan JE, Herens A, Fruchtman M, Veliz P, Kelly EL, Worster B. Cannabis use in a community-based sample of adults diagnosed with ADHD: prevalence, impact on symptoms, and stimulant side effects. *J Atten Disord*. Published online August 28, 2025. doi:10.1177/10870547251364575

88. Notzon DP, Pavlicova M, Glass A, et al. ADHD is highly prevalent in patients seeking treatment for cannabis use disorders. *J Atten Disord*. 2020;24(11):1487-1492. doi:10.1177/1087054716640109

89. Loflin M, Earleywine M, De Leo J, Hobkirk A. Subtypes of attention deficit-hyperactivity disorder (ADHD) and cannabis use. *Subst Use Misuse*. 2014;49(4):427-434. doi:10.3109/10826084.2013.841251

90. Mitchell JT, Sweitzer MM, Tunno AM, Kollins SH, McClernon FJ. "I use weed for my ADHD": a qualitative analysis of online forum discussions on cannabis use and ADHD. *PLoS One*. 2016;11(5):e0156614. doi:10.1371/journal.pone.0156614

91. Bourque J, Potvin S. Cannabis and cognitive functioning: from acute to residual effects, from randomized controlled trials to prospective designs. *Front Psychiatry*. 2021;12:596601. doi:10.3389/fpsyt.2021.596601

92. Cooper RE, Williams E, Seegobin S, Tye C, Kuntsi J, Asherson P. Cannabinoids in attention-deficit/hyperactivity disorder: a randomised-controlled trial. *Eur Neuropsychopharmacol*. 2017;27(8):795-808. doi:10.1016/j.euroneuro.2017.05.005

93. Hasin DS, Kerridge BT, Saha TD, et al. Prevalence and correlates of DSM-5 cannabis use disorder, 2012-2013: findings from the National Epidemiologic Survey on Alcohol and Related Conditions-III. *Am J Psychiatry*. 2016;173(6):588-599. doi:10.1176/appi.ajp.2015.15070907

94. Meyer C, Callahan M, Grant I. Cannabis and cognition. Accessed October 15, 2025. https://www.cannabisevidence.org/clinician-resources/clinician-briefs/cannabis-and-cognition/

95. Simmons SM, Caird JK, Sterzer F, Asbridge M. The effects of cannabis and alcohol on driving performance and driver behaviour: a systematic review and meta-analysis. *Addiction*. 2022;117(7):1843-1856. doi:10.1111/add.15770

96. Asbridge M, Hayden JA, Cartwright JL. Acute cannabis consumption and motor vehicle collision risk: systematic review of observational studies and meta-analysis. *BMJ*. 2012;344:e536. doi:10.1136/bmj.e536

97. Els C, Jackson TD, Tsuyuki RT, et al. Impact of cannabis use on road traffic collisions and safety at work: systematic review and meta-analysis. *Can J Addict*. 2019;10(1):8-15. doi:10.1097/CXA.0000000000000046

98. Wieghorst A, Roessler KK, Hendricks O, Andersen TE. The effect of medical cannabis on cognitive functions: a systematic review. *Syst Rev*. 2022;11(1):210. doi:10.1186/s13643-022-02073-5

99. Smith SA, Le GH, Teopiz KM, et al. Effects of cannabidiol and Δ^9-tetrahydrocannabinol on cytochrome P450 enzymes: a systematic review. *Drug Metab Rev*. 2024;56(2):164-174. doi:10.1080/03602532.2024.2346767

100. Penn State College of Medicine. CANN-DIR: CANNabinoid Drug Interaction Review. Accessed December 2, 2025. https://cann-dir.psu.edu/

101. Anderson LL, Absalom NL, Abelev SV, et al. Coadministered cannabidiol and clobazam: preclinical evidence for both pharmacodynamic and pharmacokinetic interactions. *Epilepsia*. 2019;60(11):2224-2234. doi:10.1111/epi.16355

102. Vaughn SE, Strawn JR, Poweleit EA, Sarangdhar M, Ramsey LB. The impact of marijuana on antidepressant treatment in adolescents: clinical and pharmacologic considerations. *J Pers Med*. 2021;11(7):615. doi:10.3390/jpm11070615

103. Wilens TE, Biederman J, Spencer TJ. Case study: adverse effects of smoking marijuana while receiving tricyclic antidepressants. *J Am Acad Child Adolesc Psychiatry*. 1997;36(1):45-48. doi:10.1097/00004583-199701000-00016

104. Vierke C, Marxen B, Boettcher M, Hiemke C, Havemann-Reinecke U. Buprenorphine-cannabis interaction in patients undergoing opioid maintenance therapy. *Eur Arch Psychiatry Clin Neurosci*. 2021;271(5):847-856. doi:10.1007/s00406-019-01091-0

105. Ridout KK, Young-Wolff KC, Ridout SJ. A change in blood carbamazepine levels associated with cannabis use: implications for clinical practice. *J Clin Psychiatry*. 2021;82(3):20cr13777. doi:10.4088/JCP.20cr13777

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106. Florian J, Salcedo P, Burkhart K, et al. Cannabidiol and liver enzyme level elevations in healthy adults: a randomized clinical trial. *JAMA Intern Med*. 2025;185(9):1070-1078. doi:10.1001/jamainternmed.2025.2366

107. Nachnani R, Knehans A, Neighbors JD, et al. Systematic review of drug-drug interactions of delta-9-tetrahydrocannabinol, cannabidiol, and *Cannabis*. *Front Pharmacol*. 2024;15:1282831. doi:10.3389/fphar.2024.1282831

108. Systematically Testing the Evidence on Marijuana. Policy map. Accessed October 27, 2025. https://www.cannabisevidence.org/policy/

109. Merlin JS, Althouse A, Feldman R, et al. Analysis of state cannabis laws and dispensary staff recommendations to adults purchasing medical cannabis. *JAMA Netw Open*. 2021;4(9):e2124511. doi:10.1001/jamanetworkopen.2021.24511

110. Bhaskar A, Bell A, Boivin M, et al. Consensus recommendations on dosing and administration of medical cannabis to treat chronic pain: results of a modified Delphi process. *J Cannabis Res*. 2021;3(1):22. doi:10.1186/s42238-021-00073-1

111. Kansagara D, Hill KP, Yost J, et al; Population Health and Medical Science Committee of the American College of Physicians. Cannabis or cannabinoids for the management of chronic noncancer pain: best practice advice from the American College of Physicians. *Ann Intern Med*. 2025;178(5):714-724. doi:10.7326/ANNALS-24-03319

112. Busse JW, Vankrunkelsven P, Zeng L, et al. Medical cannabis or cannabinoids for chronic pain: a clinical practice guideline. *BMJ*. 2021;374(2040):n2040. doi:10.1136/bmj.n2040

113. Fischer B, Robinson T, Bullen C, et al. Lower-Risk Cannabis Use Guidelines (LRCUG) for reducing health harms from non-medical cannabis use: a comprehensive evidence and recommendations update. *Int J Drug Policy*. 2022;99:103381. doi:10.1016/j.drugpo.2021.103381

114. Krist AH, Davidson KW, Mangione CM, et al; US Preventive Services Task Force. Screening for unhealthy drug use: US Preventive Services Task Force Recommendation Statement. *JAMA*. 2020;323(22):2301-2309. doi:10.1001/jama.2020.8020

115. Matson TE, Lapham GT, Bobb JF, et al. Validity of the Single-Item Screen-Cannabis (SIS-C) for cannabis use disorder screening in routine care. *JAMA Netw Open*. 2022;5(11):e2239772. doi:10.1001/jamanetworkopen.2022.39772

116. Myers MG, Ganoczy D, Walters HM, Pfeiffer PN, Ilgen MA, Bohnert KM. Assessing the diagnostic utility of the Cannabis Use Disorder Identification Test - Revised (CUDIT-R) among veterans with medical and non-medical cannabis use. *Drug Alcohol Depend*. 2023;247:109876. doi:10.1016/j.drugalcdep.2023.109876

117. Budney AJ, Borodovsky JT, Struble CA, et al. Estimating THC consumption from smoked and vaped cannabis products in an online survey of adults who use cannabis. *Cannabis Cannabinoid Res*. 2024;9(2):688-698. doi:10.1089/can.2022.0238

118. McDonagh MS, Morasco BJ, Wagner J, et al. Cannabis-based products for chronic pain: a systematic review. *Ann Intern Med*. 2022;175(8):1143-1153. doi:10.7326/M21-4520

119. Lo JO, Ayers CK, Yeddala S, et al. Prenatal cannabis use and neonatal outcomes: a systematic review and meta-analysis. *JAMA Pediatr*. 2025;179(7):738-746. doi:10.1001/jamapediatrics.2025.0689

120. ACOG Clinical Consensus No. 10: Cannabis Use During Pregnancy and Lactation. *Obstet Gynecol*. 2025;146(4):600-611. doi:10.1097/AOG.0000000000006053

121. Connor JP, Stjepanović D, Budney AJ, Le Foll B, Hall WD. Clinical management of cannabis withdrawal. *Addiction*. 2022;117(7):2075-2095. doi:10.1111/add.15743

122. Perron BE, Holt KR, Yeagley E, Ilgen M. Mental health functioning and severity of cannabis withdrawal among medical cannabis users with chronic pain. *Drug Alcohol Depend*. 2019;194:401-409. doi:10.1016/j.drugalcdep.2018.09.029

123. Coughlin LN, Ilgen MA, Jannausch M, Walton MA, Bohnert KM. Progression of cannabis withdrawal symptoms in people using medical cannabis for chronic pain. *Addiction*. 2021;116(8):2067-2075. doi:10.1111/add.15370

124. Ghasemiesfe M, Ravi D, Vali M, et al. Marijuana use, respiratory symptoms, and pulmonary function: a systematic review and meta-analysis. *Ann Intern Med*. 2018;169(2):106-115. doi:10.7326/M18-0522

125. Tashkin DP, Roth MD. Pulmonary effects of inhaled cannabis smoke. *Am J Drug Alcohol Abuse*. 2019;45(6):596-609. doi:10.1080/00952990.2019.1627366

126. Gallagher TJ, Chung RS, Lin ME, Kim I, Kokot NC. Cannabis use and head and neck cancer. *JAMA Otolaryngol Head Neck Surg*. 2024;150(12):1068-1075. doi:10.1001/jamaoto.2024.2419

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